

Lecture 01: Introduction to Reinforcement Learning

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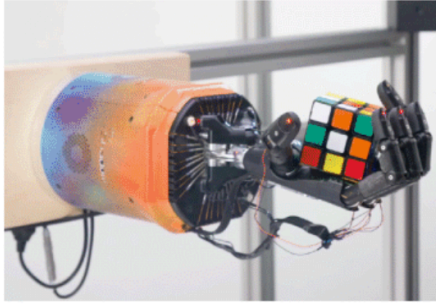
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Organizational info

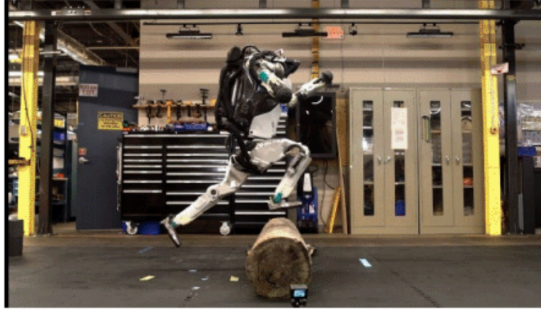
- Three homeworks
- Grading: $0.1A + 0.4HW + 0.5E$
 - A — for activity on lectures & seminars
 - HW — homeworks
 - E — exam
- Contact us via dedicated Telegram channel

What RL is about?

Where we are today?



openAI



Atlas, BostonDynamics



Amazon



TossingBot, google



Tesla self-driving car



DeepMind, AlphaGo

Examples / success stories

- games: Atari, AlphaGo/AlphaZero, Minecraft
- robotics: manipulation, locomotion, sim2real
- energy/control: data center cooling, HVAC control
- logistics: warehouse robots, order picking (multi-agent)
- recommender systems: ranking/personalization
- ops research: inventory, pricing, routing, traffic signals
- healthcare (research): dosing / treatment policies

Good behavior in any situation

or a bit less vague

Learn to act and accomplish goals in dynamic environments

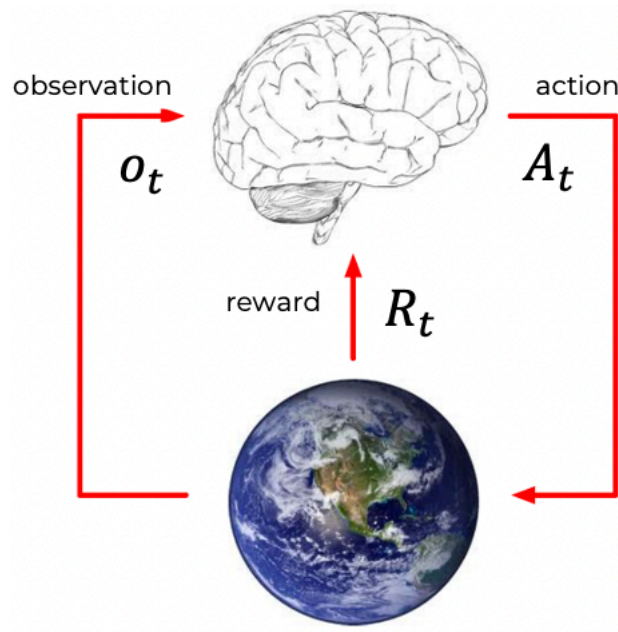
Interaction loop & core terms

Agent $\leftrightarrow$ Environment

- unit time steps: $t = 0, 1, 2, \dots$
- state/observation: s_t / o_t
- action: a_t
- reward: r_t
- next state: s_{t+1}
- trajectory: $s_0, a_0, r_0, s_1, a_1, r_1, \dots$

Policy = behavior

- deterministic: $a = \pi(s)$
- stochastic: $a \sim \pi(\cdot \mid s)$



What we care about

- Reward r_t is local, but the goal is global.
- Return G_t is a future sum:

$$G = r_0 + \gamma r_1 + \gamma^2 r_2 + \dots = \sum_{t=0}^{\infty} \gamma^t r_t$$

- or $G_t = r_t + \gamma r_{t+1} + \gamma^2 r_{t+2} + \dots = \sum_{i=t}^{\infty} \gamma^{i-t} r_i$
- Calculated for each passing state $G(s_t)$ or timestep G_t , and G_0 defines a “life” score
- Stochastic in general $\Rightarrow$ we may care about average returns
- $\gamma \in [0, 1)$ is a discount factor
 - Why? Time distance based attention, convergency
- Reward arrives later (delayed feedback) $\Rightarrow$ Which action caused it? Credit assignment
 - core RL problem

What is agent's “life”?

Episode – a life

- End via terminal states
- May have infinite length
- Often some “max episode length” is set to prevent that — episode truncation

Episodic vs continuing setting:

- episodic: terminal states, finite max length T
- continuing: no terminal, ongoing interaction

We will mostly consider episodic setting with truncation

Markov Decision Process (MDP)

MDP is a problem definition

- states S , actions A
- transition function $T(s, a) = \Pr(s' \mid s, a)$
- reward function $R(s, a, s')$
- discount factor γ
- **objective:**

$$\pi^* = \arg \max_{\pi} \mathbb{E}[G_0]$$

Markov property (intuition)

No need to collect history of agent/env interactions, state is “enough”

$$P(s_{t+1} \mid s_t, a_t, \text{history}) = P(s_{t+1} \mid s_t, a_t)$$

- future depends only on current state + action
- good state representation matters
- if not Markov $\rightarrow$ need memory (later)

MDP is not the only setting

- convenient baseline, dominates in RL
- simple enough to analyze
- rich enough for many tasks
- extensions exist:
 - bandits, multi-agent
 - partially observable MDP, semi-MDP, non-markov, differential RL

SL vs RL

Supervised learning

- data: fixed, passive
- feedback: labels, informative
- optimize prediction error
- i.i.d-ish samples
- time: often 1-step

Reinforcement learning

- data: actively collected
- feedback: reward, less informative, delayed
- optimize returns — future rewards sum
- correlated data
- time: sequences

Less supervision → harder optimization,
but easier to define the task

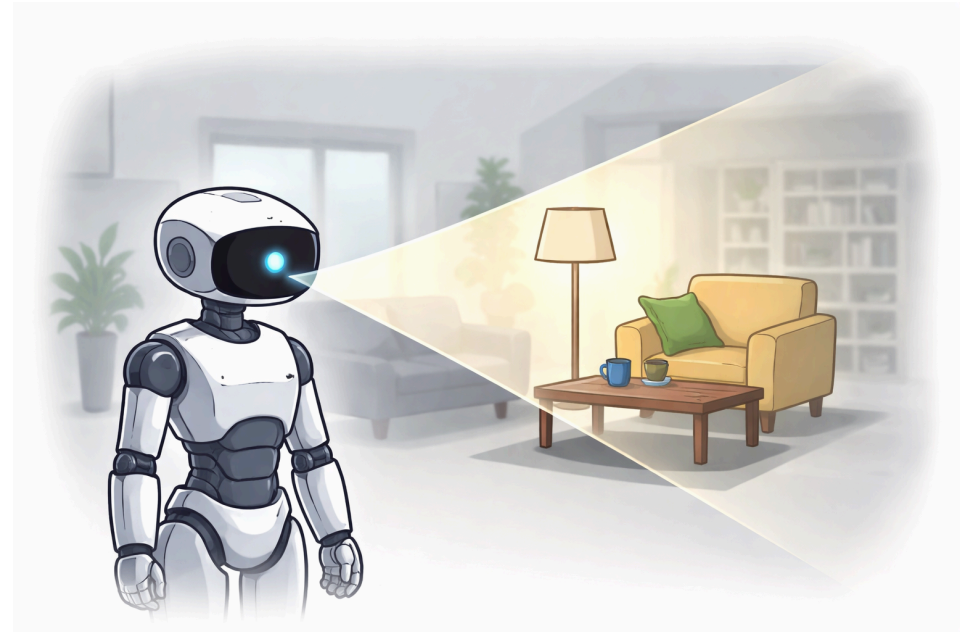
Observation vs state

Observability

- full state s_t (MDP)
- observation o_t (partial info)

If partial:

- memory / belief needed
- Partially observable MDP (POMDP)



Stochasticity

Where randomness may appear

- environment transitions $T(s'|s, a)$
- reward stochasticity $R(s, a, s')$
- stochastic policy $\pi(a|s)$

Expected return $\mathbb{E}[G(s)]$ requires an expectation over each of them

What's wrong/hard about returns

Returns are painful targets

- depend on future
- **noisy** and high variance on long horizon

Idea: predict expected return $\Rightarrow$ value functions

State value function V — “how good is state”:

$$V^\pi(s) = \mathbb{E}[G(s_t) \mid s_t = s]$$

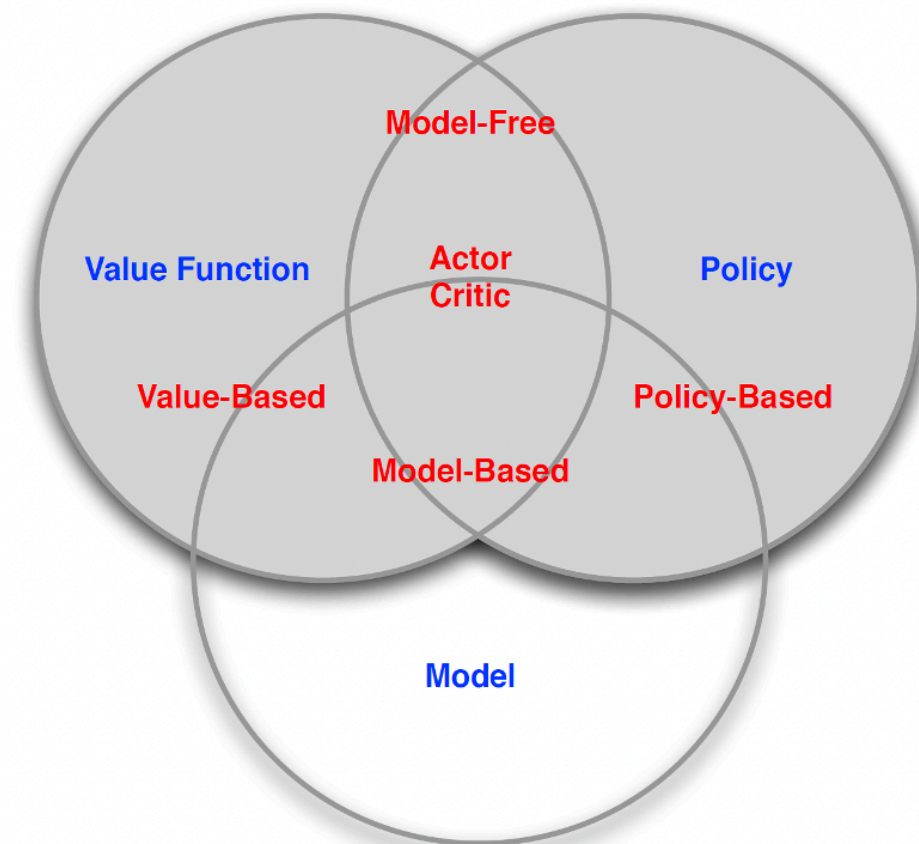
Action value function Q — “how good is action”

$$Q^\pi(s, a) = \mathbb{E}[G(s_t, a) \mid s_t = s, a_t = a]$$

Common agent components

3 objects

- policy π : choose actions
- value V/Q : estimate future
- model $M = (T, R)$: predict dynamics



Why RL is hard

rewards

- delayed reward — credit assignment
- sparse reward — rare informative signal

exploration

- huge search space
- need trials to discover good behavior
- many steps to goal

data issues (non-iid)

- correlated trajectories
- distribution shift
- bad policy → bad data loop

Exploration vs exploitation (XX dilemma)

- exploit: best-known action → reward now
- explore: try unknown/suboptimal → learn better
- no way to validate when to stop
- balance needed

SL → RL spectrum

Learning to act from data

- imitation learning: behavior cloning
- sequence prediction: Decision Transformer, implicit optimization
- offline RL: fixed dataset, reward-based
- online/hybrid RL: active interaction

Planning / control → RL

Classical control:

- model known
- design controller — reactive feedback rule (fast “policy”)

Planning:

- known fast simulator
- plan to optimize — online lookahead computation (explicit search)

MPC (Model Predictive Control):

- hybrid planning + control

Model-based RL:

- learn model
- plan or use model another way

Model-free RL:

- no explicit model

Takeaways

- RL = learn behavior over time with delayed rewards
- objective: maximize expected return
- MDP formalism
- terms: state, observation, action, reward, policy, return, value functions
- core difficulties:
 - credit assignment
 - exploration
 - non-iid data

Recommended literature / sources

- Reinforcement learning: An Introduction (Sutton, Barto) — classical theoretic book ([ru version](#))
- DeepMind x UCL | Introduction to Reinforcement Learning 2015 — nice video course on basic RL
- Deep RL theory summary (Sergey Ivanov) — modern approaches, ru book
- Deep Reinforcement Learning, a textbook (Aske Plaat)
- Reinforcement Learning: An Overview (Kevin Murphy)
- CMU Deep RL 10-403 course (Katerina Fragkiadaki)
- Yandex SDA RL course (Just Heuristic, Pavel Shvechikov, Alexander Fritsler et al.)